Frontier



Carbonlantis

Carbon dioxide removal prepurchase application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Carbonlantis

Organization location (we welcome applicants from anywhere in the world)

Konaseema, Andhra Pradesh, India

Name(s) of primary point(s) of contact for this application

Venkata Ramana Mattaparthi and Sai Durga Yerubandi

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Carbonlantis is a for-profit CDR company enabling high-rate alkalinity addition in inland waters through MRV-driven dosing and controlled near-field mixing.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Konaseema Inland Water Alkalinity Enhancement Project

[1] We use “project” throughout this template, but the term is not intended to denote a single facility. The “project” being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your organization that transforms.

b. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us how your project addresses any of the priority innovation areas identified in the RFP. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Carbonlantis is developing a first-of-a-kind Inland Water Alkalinity Enhancement (IWAE) platform for tropical, ocean-connected managed waterways. Our first Frontier project is the Konaseema Inland Water Alkalinity Enhancement Project in the Godavari delta region of Andhra Pradesh, India. The project is designed to test and deliver durable carbon removal by adding safety-screened, calcium-rich alkalinity to managed canals, creeks, drains, and distributaries that connect downstream to marine systems, effectively enhancing the alkalinity addition rates for ocean carbon sinks.

The core carbon removal mechanism is alkalinity-driven conversion of dissolved and atmospheric CO2 into stable dissolved inorganic carbon, primarily bicarbonate. Carbonlantis uses circular feedstocks, currently focusing on Calcium Carbide Residue (CCR), a Ca(OH)2-rich industrial byproduct, as the initial alkaline feedstock. Qualified CCR is screened, quarantined, and conditioned into a standardized alkaline mineral suspension in a centralized slurry refinery. The highly reactive, refined slurry is then dosed through precision pumps into managed inland waters only during suitable hydrological and carbonate-chemistry windows. The project does not assume that all theoretical alkalinity becomes credited CDR. Credited removals will be determined ex post through measured alkalinity delivery, dissolution/availability, carbonate chemistry response, downstream persistence, precipitation correction, life-cycle emissions deductions, and uncertainty discounts.

The simplified reaction pathway for calcium hydroxide-rich material is:
$Ca(OH)_2 + 2CO_2 \rightarrow Ca^{2+} + 2HCO_3^-$

In field conditions, this stoichiometry is only a starting point. Actual removal depends on real-time empirical local parameters under field conditions, such as active Ca(OH)2 fraction, particle size, receiving-water pH, DIC, total alkalinity, calcium, salinity, saturation state, mixing intensity, residence time, precipitation loss, downstream export, and MRV uncertainty. Our current feedstock basis assumes CCR with approximately 55-60% active Ca(OH)2 by mass, derived from batch lab validations. The representative refinery batch is approximately 5,000 kg CCR mixed with 50,000 L water, producing a conditioned alkaline suspension of approximately 95.66 g CCR/L, or approximately 10% w/v. We avoid describing this as a fully dissolved solution; it is a high-reactivity mineral suspension whose credited carbon removal must be validated through field chemistry and MRV.

The project architecture has three layers. First, a centralized slurry refinery receives and screens CCR, prepares standardized slurry in large HDPE mixing tanks, separates usable top-fraction slurry from reject material, and stores ready slurry for controlled dosing. Second, a centralized high-flow dosing hub deploys 12 precision dosing stations and 12 Vortex Mixing and Containment System (VMCS) units in our pilot Godavari distributary corridor. Third, a distributed microdosing network can receive slurry from the same refinery and dose smaller canals, creeks, and drains using repeatable low-CAPEX

dosing modules. The Frontier prepurchase focuses on the FOAK centralized refinery and dosing hub, while the current canal microdosing project validates the distributed unit boundary.

Carbonlantis addresses two Frontier innovation priorities directly. The first is to maximize alkalinity addition rates for OAE/IWAE while reducing the risk of secondary precipitation. The main bottleneck for alkalinity enhancement is not only sourcing alkalinity, but adding it to real waters without creating high-pH hotspots, undissolved plumes, or premature carbonate precipitation. Our VMCS is designed as a semi-contained, flow-through near-field mixing interface. It creates a repeatable dosing-zone control volume where precision-dosed slurry is mixed with upstream water before downstream release. The system is designed to improve slurry-water contact, increase local turbulence, reduce pH excursions, and generate measurable near-field data on dissolution and precipitation dynamics. We do not claim that the VMCS has already eliminated secondary precipitation; the FOAK project will empirically quantify and suppress precipitation under field conditions using filtered/unfiltered TA/DIC, calcium mass balance, turbidity, plume mapping, and saturation-state modeling. This FOAK deployment is designed to test whether controlled near-field mixing via VMCS can reduce secondary precipitation losses below ~5–10% under real IWAE field conditions, while maintaining stable downstream alkalinity persistence.

The second Frontier priority is improving MRV for open-system pathways. Carbonlantis treats MRV as an operating control layer, not only as a reporting layer. Our three-station architecture includes: an upstream predictive gate to measure baseline chemistry and determine whether dosing is allowed; a dosing-zone safety station to verify delivery, pH response, plume behavior, and stop triggers; and a downstream crediting station to measure alkalinity persistence and support carbon accounting. Continuous sensors such as pH, EC, temperature, turbidity, water level, and flow support operational control, while MRV-grade accounting depends on total alkalinity, DIC, pCO2, calcium, major ions, filtered/unfiltered samples, and discharge estimates. This layered architecture is intended to improve the signal-to-noise ratio in managed waterways with defined geometry compared with fully open-ocean settings. Beyond project-level quantification, this architecture is intended to enable a transition from site-specific MRV toward cluster-level, model-driven MRV frameworks, reducing long-run monitoring cost while maintaining high-confidence attribution.

Carbonlantis is differentiated from other alkalinity approaches in five ways. First, we focus on managed tropical inland waters rather than only on open-ocean or coastal outfall settings. These systems are not closed reactors, but they provide more defined monitoring corridors and operational control volumes. Second, we use a centralized refinery plus a distributed dosing architecture, avoiding the need to replicate full processing infrastructure at every dosing site. Third, the VMCS provides an engineered near-field mixing interface designed specifically around slurry dosing and precipitation risk. Fourth, the system is responsive: dosing can be reduced, paused, or adjusted based on local pH, flow, saturation state, turbidity, and downstream signal. Fifth, our first feedstock is circular: CCR is an industrial residue that can reduce reliance on virgin-mined alkalinity if its safety, additionality, and leachability are validated.

The durability pathway is downstream persistence of added alkalinity and carbon as dissolved inorganic carbon/bicarbonate through the river-estuary-ocean continuum. Bicarbonate storage in the ocean is generally understood to occur over millennial-to-multi-millennial timescales, but the project

must still quantify local losses due to incomplete dissolution, secondary precipitation, degassing, and background variability. For this reason, Carbonlantis is not presenting IWAE as a solved pathway. We are presenting a field-ready, milestone-based deployment designed to generate transferable evidence for near-field mixing, secondary precipitation thresholds, and scalable MRV in tropical managed waterways.

If successful, Carbonlantis can expand from the FOAK Godavari distributary hub into a district-level network of centralized refineries supplying high-flow dosing hubs and distributed canal/creek microdosing sites. The same architecture can later be adapted for wastewater treatment plants, high-decay urban municipal drains, and eventually contained alkalinity-contacting systems for BioCCS, BECCS, DACCS, and other industrial CO2-bearing fluids. Frontier support would therefore buy early net tonnes and accelerate the emergence of a new field category: measurable, modular, inland-water alkalinity enhancement in the Global South.

c. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics, and be specific but concise. 1000-1500 words

Carbonlantis is trying to build the first integrated FOAK Inland Water Alkalinity Enhancement system that can safely, measurably, and repeatedly deliver net carbon removal from managed inland water networks. The project is located in the Konaseema district, within the Godavari delta region of Andhra Pradesh, India, with the FOAK focused on a centralized slurry refinery and a centralized high-flow dosing hub near a Godavari distributary corridor. The current canal microdosing unit serves as a validation layer for distributed dosing modules supplied by the same centralized slurry-preparation model.

The main engineering objective is to validate the complete refinery-dosing-MRV chain. A key objective is to define the operational envelope for alkalinity addition, including validated thresholds for precipitation loss, dosing density, and downstream persistence under varying hydrological and carbonate chemistry conditions. The refinery will receive CCR, screen and quarantine feedstock, prepare a standardized alkaline slurry, separate usable ready slurry from reject fractions, and supply the dosing hub. The FOAK dosing hub will use 12 precision dosing stations, 12 VMCS units, upstream/dosing/downstream MRV, and PLC/cloud feedback controls. A current-project microdosing unit will demonstrate the repeatable distributed site boundary: slurry supplied from centralized preparation, local buffer tanks, 6 peristaltic dosing lines, 6 VMCS/local mixing units, upstream/dosing/downstream monitoring, solar/battery power, field O&M, and local refill logistics.

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

The project scale in the Frontier TEA is intentionally conservative. The FOAK has a gross nameplate capacity of 3,000 tCO2/year. It is modeled at a 60% capacity factor, equivalent to approximately 5,256 operating hours per year, which is within the expected 4,800-5,400-hour annual suitable dosing window. On this basis, the FOAK produces 1,800 tCO2/year of gross active uptake and deducts 300 tCO2e/year of life-cycle emissions, resulting in 1,500 tCO2/year of net removal. The proposed Frontier offer is 1,800 net tons delivered over 18-24 months at $780/tCO2, with the 24-month case treated as the base delivery case and the 18-month case as an upside case if commissioning and hydrology proceed faster than modeled.

The cost objective is to prove a credible path from FOAK cost to mature cost. The final TEA separates three stages. The current microdosing project is approximately $214/tCO2 all-in, including MRV, but it excludes full refinery CAPEX because it receives conditioned slurry from a centralized preparation/refinery point. The FOAK is approximately $601/tCO2 all-in, including MRV; the Cost Summary tab reports approximately $536/tCO2 because the template summary excludes MRV, while the Cost Inputs row 88 (TEA) is treated as the all-in cost. The NOAK template-derived value is approximately $95/tCO2 before any separately modeled mature MRV sensitivity; adding $5-10/tCO2 for mature cluster-level MRV yields an all-in range near $100-105/tCO2.

The largest current FOAK cost components are levelized CapEx, fixed OPEX, and MRV. FOAK levelized CapEx is approximately $326/tCO2, driven by a conservative Total Overnight Cost of approximately $1.78M that includes bare erected cost, EPC, TRL 6 process contingency, Class-I project contingency, and owner's costs. Fixed OPEX is approximately $180/tCO2 and captures site operations, maintenance, security, compliance, field troubleshooting, supplier qualification, and early operating burden. MRV is approximately $65/tCO2, intentionally elevated because first deployments in open and semi-open water systems must overmeasure to reduce uncertainty. Other variable OPEX is intentionally modeled as a marginal, throughput-linked cost only, at approximately $24/gross tCO2 or $28.8/net tCO2, covering regional logistics, consumables/spares, QA/reject handling, and feedstock/refinery handling. FOAK-specific inefficiencies are captured in fixed OPEX, MRV, conservative utilization, and Frontier-template contingencies rather than hidden in variable OPEX.

To meet Frontier's long-run cost and scale criteria, five conditions must hold. First, centralized slurry refinery systems must become standardized regional infrastructure that supplies multiple dosing nodes rather than one-off custom plants. Second, VMCS and dosing platforms must become repeatable modules with lower procurement, installation, cleaning, and maintenance costs. Third, feedstock sourcing must mature through regional supplier qualification, lower logistics distances, and consistent batch QA. Fourth, MRV must compress from dense FOAK sampling to a cluster-level architecture that combines representative high-grade stations, lower-cost distributed sensors, and model-centric uncertainty treatment. Fifth, permitting and community engagement must become repeatable, transparent, and trusted across managed waterway deployments.

Quantification follows a conservative carbon accounting ladder. Gross theoretical potential is not credited directly. The project will measure or model the following: feedstock active alkalinity, slurry concentration, delivered volume, dissolution/availability, precipitation loss, downstream persistence, background variability, life-cycle emissions, and uncertainty. Upstream stations establish baseline chemistry and suitability gates. Dosing-zone stations confirm delivery and safety. Downstream stations

measure persistence and support crediting. Continuous pH/EC/pCO2/temperature/turbidity/flow data support control, while TA, DIC, saturation state, calcium, major ions, filtered/unfiltered samples, and laboratory validation support CDR accounting.

The project will also produce field-learning value beyond its first tonnes. The most important knowledge outputs are: precipitation-loss factors under real IWAE dosing conditions; pH and alkalinity gradients across the near-field mixing zone; VMCS mixing and residence-time performance; feedstock batch variability and release criteria; and a practical MRV architecture for managed inland waters. These are directly aligned with Frontier's priorities around alkalinity addition rates, secondary precipitation, and MRV for open-system pathways.

Carbonlantis does not present the first FOAK as the final optimized design. It is a deliberately instrumented, risk-managed deployment. The operating philosophy is to prioritize safe, measurable, verifiable dosing over maximum throughput. If field data shows higher precipitation, lower export, feedstock variability, or community concerns, the system will throttle, pause, rework feedstock, or restrict crediting until evidence supports delivery. This is the reason the offer volume is smaller than the full five-year FOAK modeled output and why the price includes a risk buffer over the all-in FOAK TEA cost.

At scale, Carbonlantis aims to deploy district-level networks across the Godavari delta and similar tropical deltas. The near-term focus is on managed canals, creeks, drains, and distributaries. The 2030+ focus includes wastewater treatment plants and high-decay municipal drains where alkalinity addition can combine CDR with water-conditioning benefits, while keeping any avoided emissions outside credited CDR unless allowed by protocol. Post-2035, the same high-shear alkalinity-contacting logic may be adapted into contained reactors for BioCCS, BECCS, DACCS, and industrial CO2-bearing fluids. The Frontier project is therefore both an early CDR delivery and a step toward a broader category of alkalinity-enhancement infrastructure.

d. **Risks:** What are the biggest risks and how will you mitigate those? Include technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. 500-1000 words

Carbonlantis' largest risks are technical, MRV-related, execution-related, ecosystem-related, governance-related, and financial. We treat these risks as core pilot objectives rather than side issues.

Technical risk: the key risk is that added alkalinity may not translate into durable net CDR at the assumed efficiency due to incomplete dissolution, secondary precipitation, plume behavior, or receiving-water chemistry. We mitigate this through feedstock QA, slurry conditioning, VMCS near-field mixing, saturation-state-responsive dosing, PHREEQC/carbonate chemistry modeling, filtered/unfiltered TA/DIC, calcium mass balance, turbidity monitoring, and conservative uncertainty discounts. We will not claim theoretical feedstock potential as credited removal unless field data support dissolution, persistence, and net negativity.

Feedstock risk: CCR can vary by supplier, moisture content, carbonation state, active Ca(OH)2 fraction, contaminants, and particle behavior. We mitigate this through supplier qualification, batch IDs, chain-of-custody records, XRD, ICP-MS, leachate testing, active alkalinity titration, moisture testing, solids fraction checks, and a two-gate release process: raw CCR intake and refined slurry release. Lots that fail active alkalinity, contaminant, or leachability criteria will be rejected, redirected to non-CDR handling, or treated as non-creditable material.

MRV risk: managed inland waters are variable systems. Natural pH, flow, salinity, DIC, TA, turbidity, and biological activity can obscure the project signal. Carbonlantis mitigates this through three-station MRV: upstream predictive gate, dosing-zone safety station, and downstream crediting station. We use continuous sensors for operational control and lab-anchored carbonate chemistry for accounting. Baseline periods, controlled dosing events, replicate samples, filtered/unfiltered comparisons, and uncertainty budgets will be used to avoid over-crediting.

Ecosystem and water-quality risk: high-pH slurry could create local pH spikes, turbidity, visible precipitation, or biotic stress if poorly mixed or dosed under unsuitable conditions. We mitigate this through microdosing, VMCS mixing, pH/turbidity stop-triggering, dosing only during suitable hydrological and chemical windows, local aquatic observation, downstream compliance monitoring, and immediate pause/reduction procedures. The project prioritizes safe and measurable dosing over maximum throughput.

Project execution risk: the FOAK combines slurry handling, tanks, pumps, motors, sensors, public water deployment, field staff, permitting, and data systems. Failure modes include pump clogging, tubing wear, mixer fouling, sensor drift, power interruption, delayed procurement, and field access disruptions. We mitigate these through redundant spares, preventive maintenance, VFD/flow monitoring, calibration SOPs, local operators, modular hardware, buffer storage, solar plus backup power, and a commissioning phase before full delivery.

Regulatory and governance risk: public water dosing may be misunderstood or delayed by unclear permitting pathways. Carbonlantis will secure site access, waterway/canal permissions, environmental consents or NOCs where required, Pollution Control Board permissions where applicable, transport compliance, and local panchayat/municipal approvals before commercial operations. We will maintain public-facing communication, signage, grievance channels, and engagement records.

Financial risk: FOAK costs may exceed the base TEA due to logistics, hardware, MRV, or delays. The proposed price of $780/tCO2 provides a buffer over the all-in FOAK TEA cost of approximately $601/tCO2. The offered volume of 1,800 net tons is below the five-year FOAK modeled output and is scheduled over 18-24 months to allow commissioning, ramp-up, and conservative delivery.

Durability risk: bicarbonate can persist on long timescales, but local losses from precipitation, degassing, or export uncertainty must be accounted for. Our MRV plan explicitly quantifies these factors and applies ex-post discounts. Carbonlantis will avoid claiming avoided degassing, lime displacement, or any third-party CDR in gross or net CDR unless a future accepted protocol allows it.

e. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(Should be net volume after taking into account the uncertainty discount proposed in 5c)*	1,800 net metric tons CO2. This is the offered Frontier volume after life-cycle deductions and any ex-post uncertainty discounts applied at crediting.
Delivery window *(At what point should Frontier consider your contract complete? Should match 2f)*	18-24 months from FOAK commissioning / contract start. Frontier should consider the contract complete when 1,800 net tons are delivered and documented. The base schedule is 24 months, and 18 months is an upside case.
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	$600.70/tCO2 all-in FOAK cost, including MRV, based on Cost Inputs row 88 in the final TEA. Cost Summary reports approximately $535.70/tCO2 because it excludes MRV.
Levelized price ($/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	$780/tCO2. Total proposed contract value: $1.404 M.

[3] This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).